

Education

University of Utah
B.S. Computer Science

Salt Lake City, UT
August 2020 – May 2025

Relevant Coursework: Computer Systems, Machine Learning, Computer Graphics, Algorithms, Software Practice I & II, Database Systems, Computer Networks, Foundations of Data Analysis, Models of Computation, and Linear Algebra.

Skills

Languages: Python, TypeScript, C++, SQL, Java

Software: Pandas, NumPy, React, Node.js, AWS, Azure, Docker, Next.js, .NET, REST, Django, MongoDB, Linux

Experience

L3Harris Technologies
Associate Software Engineer

Dallas, TX
June 2025 – Present

- Improved data pipeline reliability for defense sensor ingestion by engineering containerized Python and C++ microservices on AWS that normalized high-frequency data from heterogeneous hardware sources into a consistent, queryable foundation across multiple production deployments.
- Reduced deployment risk across embedded government systems by building automated CI/CD infrastructure with static analysis, integration testing, and audit trails, enabling validated releases under strict defense compliance requirements.
- Increased engineering velocity for cross-functional teams by designing modular service abstractions that decoupled hardware-specific logic from shared infrastructure, allowing hardware and firmware teams to iterate independently without breaking upstream consumers.

Doxy.me
Software Engineer

Charleston, SC
August 2024 – May 2025

- Built production AI workflow infrastructure in Python and TypeScript on AWS SageMaker and Fargate, implementing multi-step inference pipelines with confidence-based human-in-the-loop routing that determined when to surface model output versus escalate to a clinician, meeting HIPAA-compliant data handling requirements throughout.
- Replaced manual clinical data workflows by building LLM pipelines (PyTorch, Llama) and WebRTC integrations that ingested unstructured session data and produced structured, auditable outputs consumed by downstream analytics and provider-facing dashboards.
- Enabled reproducible, zero-downtime releases across multiple production environments by owning YAML-based CI/CD configuration and cloud infrastructure automation, maintaining full deployment auditability required for regulated healthcare workloads.

Projects

Telehealth Mental Health Diagnosis Tool (Doxy.me): Built an AI system that analyzes visual, speech, and tonal cues from telehealth sessions using PyTorch and Llama, delivering structured mental health insights to treating physicians via SageMaker and Fargate inference pipelines. Improved diagnostic accuracy using reinforcement learning and WebRTC video integration.

University Course Planning/Review Platform (Class Best): Built a multi-service data platform enabling student schedule matching, course overlap detection, and behavioral insights. Developed ETL pipelines in Python to process thousands of course records, user preferences, and time-series activity logs. Built a responsive frontend in React + Tailwind with interactive dashboards, heatmaps, and calendar visualizations.

Configuration Fuzzer: Developed a configuration fuzzing tool for OSS-Fuzz, automating build generation to identify critical compile-time configurations and improve code coverage, uncovering previously untested code paths and increasing bug detection effectiveness.